TwinCheck: Project Summary

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TwinCheck is a pick-and-sort system for the FANUC ER-4iA that looks before every pick and checks after it, using an AI trained only in simulation. On the real cell, on the same 10 setups, it sorted **9/10** rollouts where a traditional taught program sorted **0/10** (organizers' official scores, Oct 30, 2026).

The problem we solve

Taught robot programs fail silently when parts are not exactly where they were taught. On the real cell the 12 mm jaws leave about ±3 mm around a 10 mm cube, while hand-placed parts sat 5–33 mm and bins 1–3 cm from their planned spots. The robot grabs air, knocks the part, or crashes, and nobody notices until a bin is empty or the cell stops.

- **High-mix, low-volume lines:** every new variant means re-teaching positions, typically days of an expert's time.
- **Legacy robots:** plants run equipment up to ~40 years old and need help that bolts on without reprogramming.
- **Cost:** defects cost US discrete manufacturers an estimated $32–58B a year (NIST); about 3% of manufacturers used AI for inspection (2020 Annual Business Survey).

How it uses physical AI

TwinCheck closes the loop between a digital twin, a camera-trained AI and a real industrial arm. The only thing that crosses from simulation to reality is a 5.7 MB model file.

1. **Digital twin (offline):** NVIDIA Isaac Sim 6.0 runs the organizers' FANUC ER-4iA cell with cameras placed by their calibration, so the twin sees what the real camera sees.
2. **Perception trained in simulation:** an Ultralytics YOLO11n-OBB detector (2.66 M parameters) finds cubes, M6 screws and bins with their angles; it never saw a real training image.
3. **Act and check on the real arm:** a Python policy on a laptop GPU receives the camera image and arm pose 10 times a second, aims at where each part really is, waits for the arm to arrive, and checks three times: part there, part held, part in the right bin. A missed grasp is retried; anything else raises an alert.
4. **Three adoption levels:** Baseline (today's taught program), Watch (bolt-on alerts, no reprogramming) and Guide (camera-guided picks with retries).

Data we generated and collected

All training data was generated in simulation; real data was used only to test and diagnose, as the hackathon requires.

Data	Amount	Source	Use
Synthetic training images (v1 → v4)	500 → 2,000 → +2,000 → 3,000 images; v4 ≈ 12,000 labels	Isaac Sim + Omniverse Replicator; labels projected from 3D geometry	Training only
Randomization	Part pose and colour, bins anywhere, arm pose, lighting, clutter, distractors, camera jitter, sensor noise	Replicator	Sim-to-real robustness
Held-out simulator exams	100 images per exam	Isaac Sim, calibrated camera	Accuracy testing
Twin closed-loop rollouts	20/20 sort test, plus a moved-bin test	Our policy driving the twin	Validating every fix before robot time
Real cell recordings	7 runs, 73 recorded rollouts: top, wrist, side video + arm state at 10 Hz	Organizers' LeRobotDataset recordings	Test and diagnosis only
Our run logs	Every check, with arm pose, detections and the camera frame	TwinCheck policy	Evidence and diagnosis

The v4 detector trained for 60 epochs at 1280 px (AdamW, batch 16) on an NVIDIA DGX Spark in about 1.7 hours.

How we used the CSI equipment

We ran TwinCheck on the real FANUC ER-4iA cell through the organizers' supervised evaluation platform, and trained and simulated on an NVIDIA DGX Spark in the CSI GPU lab. The cell had an SMC MHZL2-16D2 gripper (12 mm jaw opening), top, wrist and side cameras, 10 mm cubes, M6 screws and 50 mm bins; an operator set up and scored every rollout.

Date	Run	What it showed
Oct 30	Guide v6 vs blind baseline, same 10 layouts	**9/10 vs 0/10**, the official comparison
Oct 30	Guide v5, blind baseline v1	Found the wrist winding its air hoses to its −360° limit; fixed with a grasp-angle band and a safety guard
Oct 29	Guide v4: 2 cubes + 1 screw, two bins	First full sorts: 8/10 first try; finger direction confirmed
Oct 28	Guide v3: 3 cubes	Picks worked; cubes missed hand-placed bins, so we taught the AI to see bins
Oct 23	Blind dry run	0/14: the real arm lags and cuts corners, so we made it wait for arrival before each step

Every real failure was diagnosed from the cell's recordings, reproduced in the twin, fixed and tested there before the next slot.

Evidence that it solves the problem

On the same 10 setups (2 cubes + 1 M6 screw into two bins, parts placed up to 3 cm off the taught spots), TwinCheck succeeded 9 times and the taught program never did. Scores are the organizers' operator scores.

Measure	Blind taught program	TwinCheck Guide
Rollouts fully sorted	0 / 10	**9 / 10**
95% confidence interval (Wilson)	0–28%	**60–98%**
Controller faults	7	**1**
Operator interventions	0	0
Average time per 3-part sort	—	96.5 s

- **Failures predicted:** running alongside the blind program (Watch mode), TwinCheck flagged all 20 of its picks as "will miss" before the arm moved; 19 did.
- **Simulation predicted reality:** the twin reproduced the real bin failure (2/5), proved the fix (5/5, then 20/20), and the same software scored 9/10 on the real cell.
- **Verifiable:** runs r20261031-79333b (TwinCheck) and r20261030-dcd07a (baseline) on hackathon.sidclab.org; rollout 1 of each is shown as synchronized top, wrist and side video with TwinCheck's live log.
- **Limits:** 10 rollouts per condition is a small sample; the arm is speed-capped (~55 mm/s); parts stacked in one bin can hide each other from the camera; only parts that fit the 12 mm jaws were tested.